

Shareholder activism, takeovers, and managerial discipline

Francesco Celentano

University of Lausanne and SFI

Oliver Levine

University of Wisconsin-Madison

DSE Dynamic Finance and Banking Conference

The role of activism's intervention

Managers and boards may fail to maximize shareholder value due to **misaligned incentives**

Agency conflicts are costly but present an **opportunity** for potential acquirers and activists

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Both **takeovers and activism** can **mitigate** these problems $\Leftrightarrow$ **Substitutes**

Manne (1965), Grossman & Hart (1980), Jensen (1986), Admati et al (1994), Maug (1998), DeMarzo & Urošević (2006), Back et al (2018), Gryglewicz et al (2025) etc

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Yet, **activists** may also **facilitate takeovers** $\Leftrightarrow$ **Complements**

Shleifer & Vishny (1986), Burkart (1995), Burkart et al (2000), Corum & Levit (2019), Burkart & Lee (2022), Gorbenko & Malenko (2024), Burkart et al (2024) etc

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Question

What role does activist intervention play in overcoming agency conflicts and in takeovers?

→ To what extent activism and takeovers are substitutes and complements is an open question

Denes, Karpoff & McWilliams (survey, 2017)

The role of the threat of activism

The **influence** of activism and takeovers can **extend beyond direct intervention**

The threat of being targeted can incentivize managers and boards

Takeovers: Stein (1988), Agrawal & Knoeber (1998), Novaes (2002), Chemla (2005), Lal & Miller (2015), Wang and Wu (2020) etc

Activism: Brav et al (2008), Klein & Zur (2009), Gantchev (2013), Fos (2016), Gantchev et al (2018), Johnson & Swern (2021), Albuquerque et al (2022) etc

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If the **likelihood** of activism and/or takeover is **high**, managers and boards

- Perceive a **heightened risk** of losing their position
- **Act** to avoid replacement or loss of control

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Question

How much does the threat of activism impact managerial incentives and shareholder value?

→ Isolating the effect of the threat of activism on outcomes requires a model

Brav, Jiang & Li (survey, 2022)

Outline

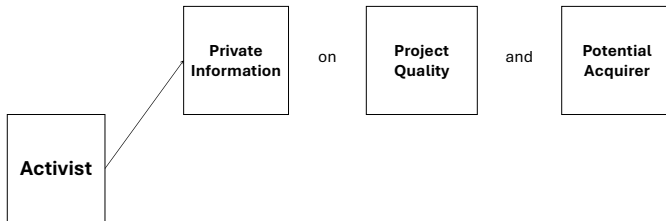
We develop a model featuring activism and takeovers



Activist

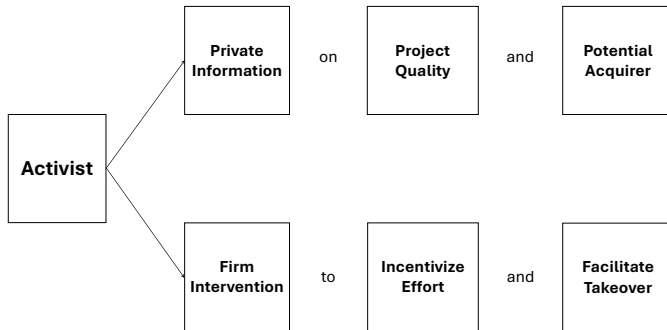
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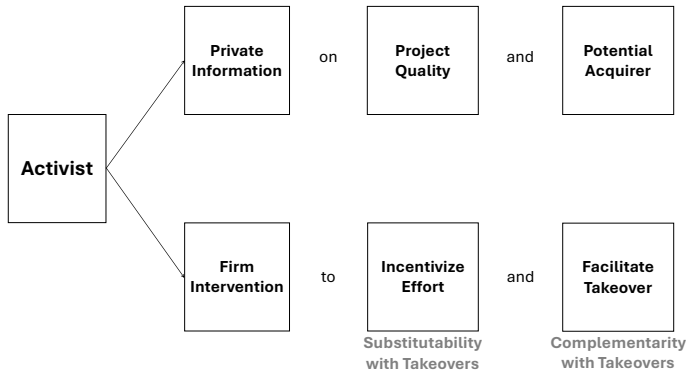
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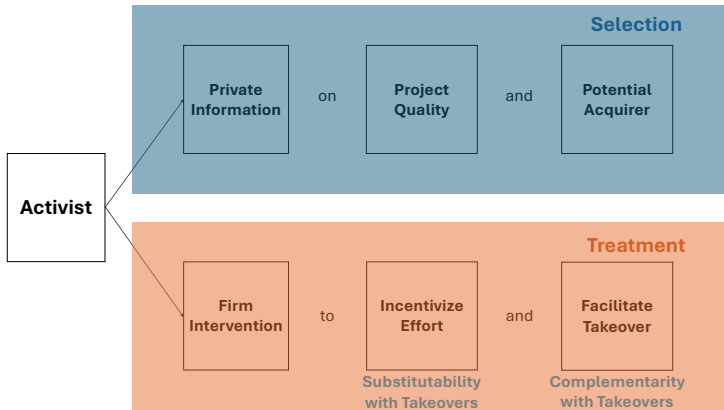
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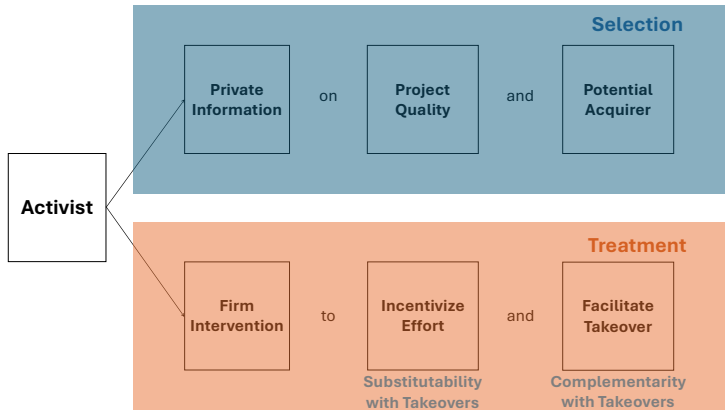
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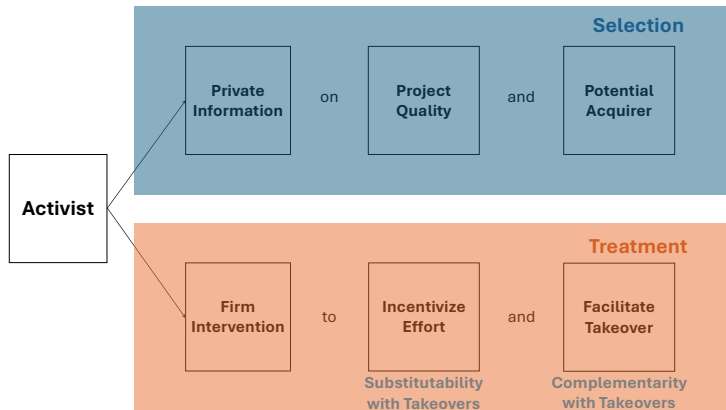
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Estimate the model to match moments on activism, takeovers, turnovers and performance

Outline

We develop a model featuring activism and takeovers



Estimate the model to match moments on activism, takeovers, turnovers and performance

Run counterfactuals to quantify the role of activism and takeovers

Roadmap

Model

Estimation

Counterfactuals

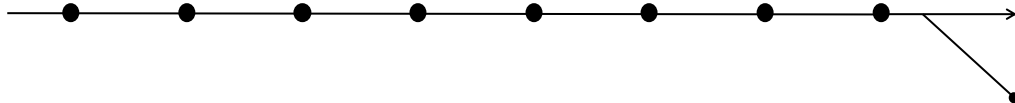
Roadmap

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Estimation

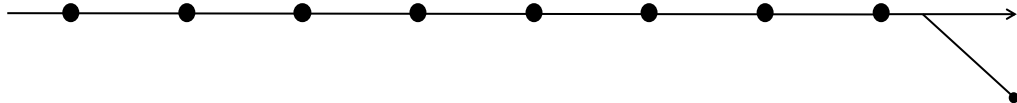
Counterfactuals

Model



Discrete time, infinite horizon, risk-neutral agents with discount rate r

Model

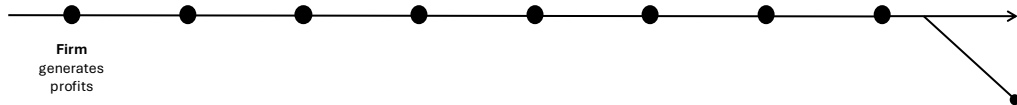


Discrete time, infinite horizon, risk-neutral agents with discount rate r

The model features

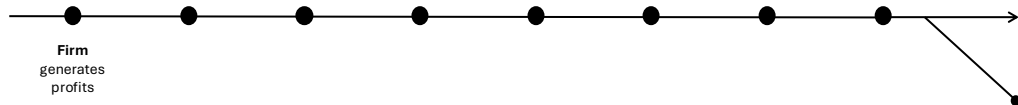
- A **firm** led by a **manager** and a **board**
- **Activist** investors
- Potential **acquirers**

Model



The **firm** generates **profits** $\pi_t = z_t k_t^\alpha - c_k k_t$

Model



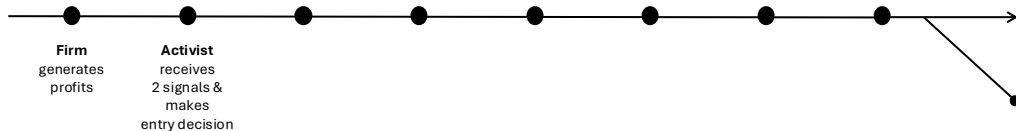
The **firm** generates **profits** $\pi_t = z_t k_t^\alpha - c_k k_t$

The **TFP** evolves as $\exp\{\log(z_{t+1})\} = \exp\{(1 - \rho_z)\mu_z + \rho_z \log(z_t) + \sigma_z \varepsilon_{t+1}\} + \underbrace{e_t p_t}_{\text{Project impact}}$

where

- e_t equals 1 if the **manager exerts effort** and 0 otherwise
- $p_t \sim \text{Exp}(\lambda_p)$ captures the **project quality**

Model

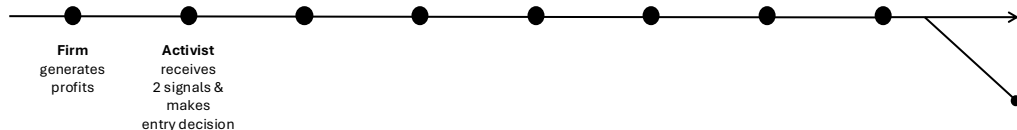


An **activist** investor observes

1) A signal about the project quality

$$s_{t,p} = p_t + \sigma_p \eta_{t,p}$$

Model



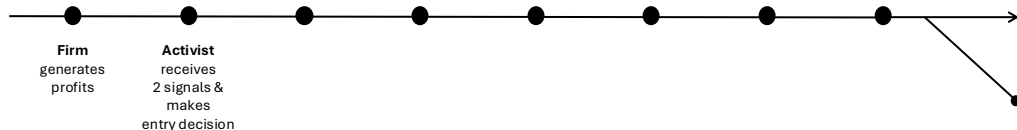
An **activist** investor observes

- 1) A signal about the project quality $s_{t,p} = p_t + \sigma_p \eta_{t,p}$
- 2) A signal about a potential acquirer $s_{t,w} = w_t + \sigma_w \eta_{t,w}$

where

- $w_t \sim \text{Exp}(\lambda_w)$ is the **acquirer quality** level
- σ_i with $i = p, w$ captures the degree of the **signal's noise**
- $\eta_{t,i}$ with $i = p, w$ is an iid standard normal **shock**

Model



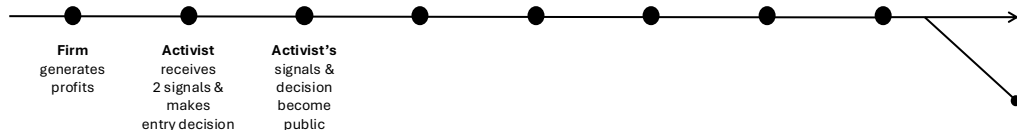
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- | | | |
|--|---------------------------------------|-------------|
| 1) A signal about the project quality | $s_{t,p} = p_t + \sigma_p \eta_{t,p}$ | } Selection |
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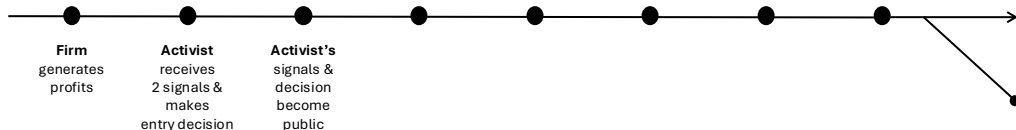
Based on this info, the **activist** makes its **entry decision** by choosing its ownership stake v

$$\max_{v \in \{0, \bar{v}\}} \underbrace{\frac{\mathbb{E}_t[V(z_{t+1})]}{1+r} v}_{\text{Expected exit value}} - \underbrace{\left((1 + \xi)(V(z_t) - \pi_t)v \right)}_{\text{Entry cost}}$$

where

- $V(\cdot)$ is the **firm value**
- ξ captures the **activist's trading and campaign costs**

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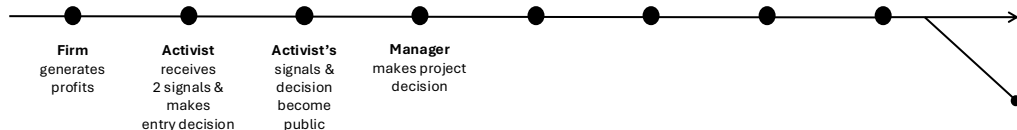
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After the activist makes its decision, its **presence and information** become **public** information

Model



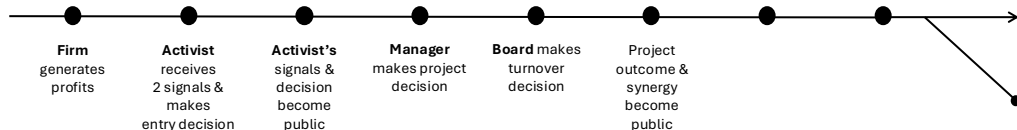
The manager decides to undertake the project or not by solving

$$M(z_t) = \max_{e_t} \theta_d \pi_t + e_t \underbrace{\left[\frac{\mathbb{E}[(1-f_t)(1-a_t)M(z_{t+1})|e_t=1]}{1+r} - c_e \pi_t \right]}_{\text{Value if the manager exerts effort}} + (1-e_t) \underbrace{\frac{\mathbb{E}[(1-f_t)(1-a_t)M(z_{t+1})|e_t=0]}{1+r}}_{\text{Value if the manager does not exert effort}}$$

where

- $\theta_d \in (0,1)$ is the fraction of the profits π_t paid to the manager
- $c_e \pi_t$ is the manager's private cost of exerting effort
- f_t is an indicator that equals 1 if a turnover occurs and 0 otherwise
- a_t is an indicator that equals 1 if a takeover occurs and 0 otherwise

Model



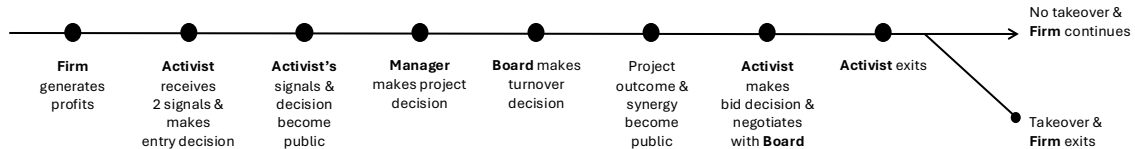
The **board** decides whether to **replace the manager or not** by solving

$$\max_{f_t} f_t \underbrace{\left[-c_f(v)\pi_t + \frac{\mathbb{E}[V(z_{t+1})|e_t=1]}{1+r} \right]}_{\text{Value if the manager is replaced}} + (1-f_t) \underbrace{\frac{\mathbb{E}[V(z_{t+1})|e_t]}{1+r}}_{\text{Value if the manager is not replaced}}$$

where

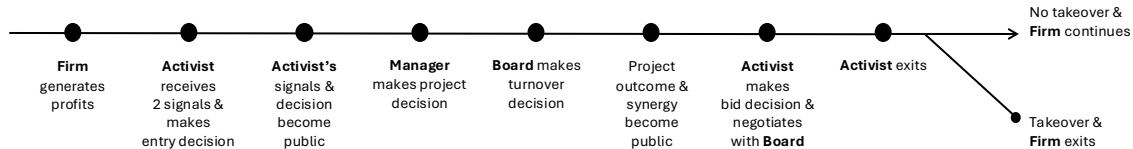
- $c_f(v)\pi_t$ is the **cost of replacing the manager** } **Treatment-Substitutability**
- $V(\cdot)$ is the **firm value**
- If a **turnover** occurs, the board appoints a new manager who **exerts effort**

Model



After the project outcome is realized and so z_{t+1} , the **acquirer quality** w_t is revealed

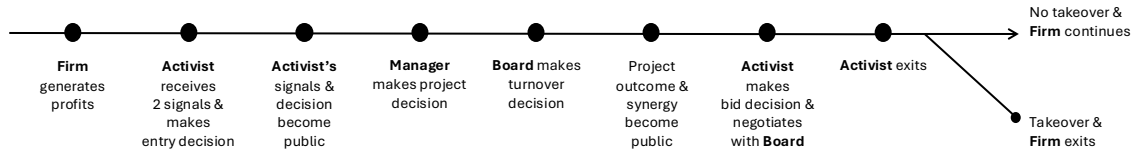
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If a **takeover** occurs, the **acquirer transfers the technology** to the target firm: $z_{t+1} \rightarrow w_t$

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If a **takeover** occurs, the **acquirer transfers the technology** to the target firm: $z_{t+1} \rightarrow w_t$

The board and the acquirer negotiate via **Nash bargaining**

The **board's utility** is
$$U(z_{t+1}) = \begin{cases} V(z_{t+1}) & \text{if there is no takeover} \\ P(1 - c_m(v)) & \text{if a takeover takes place} \end{cases}$$

where

- $V(\cdot)$ is the **firm value** while P is the **bid price**
- $c_m(v)$ reflects **entrenchment** } **Treatment-Complementarity**

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Data

We assemble a **panel** from **2001 to 2019** using

- Audit Analytics** data on activism campaigns

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We pre-set the discount rate and the productivity function parameters at standard values

We also pre-set the activist's stake $\bar{v}$ at 5%, the SEC 13D form threshold

SMM Estimation Results

Category	Description	Model	Data	t-stat
Productivity	Persistence of good performance	0.727	0.717	-0.431
	Persistence of bad performance	0.643	0.637	-0.331
	Variance of residuals	0.135	0.140	0.576
Activism	Frequency of activist entry	0.039	0.040	0.203
	Mean return on activist entry	0.041	0.044	0.629
	Coefficient of future performance on activism	0.100	0.080	-0.657
Takeovers	Frequency of takeover	0.042	0.042	-0.454
	Frequency of takeover activist	0.127	0.120	-0.887
	Mean bid premium	0.367	0.366	-0.211
	Mean bid premium activist	0.331	0.331	-0.404
Turnovers	Frequency of turnover	0.085	0.085	-0.006
	Frequency of turnover activist	0.190	0.183	-0.732
Correlations	Correlation of activism and lagged performance	-0.014	-0.017	-0.476
	Correlation of turnover and lagged performance	-0.053	-0.030	4.123
	Correlation of takeover and lagged performance	-0.033	-0.018	2.503

The model matches quite well all the 15 targeted moments

The only exceptions are provided by two of the correlations

Estimated parameters via SMM

Symbol	Description	Value	SE
ρ_z	Productivity persistence	0.860	(0.008)
σ_z	Productivity volatility	0.182	(0.010)
λ_w	Acquirer quality distribution rate	2.285	(0.056)
σ_w	Acquirer quality signal noise	2.438	(0.264)
λ_p	Project quality distribution rate	11.909	(1.171)
σ_p	Project quality signal noise	0.220	(0.043)
c_e	Manager's cost of effort	0.740	(0.271)
$c_f(O)$	Cost of turnover without activist	2.072	(0.331)
$c_f(\bar{v})$	Cost of turnover with activist	1.618	(0.296)
$c_m(O)$	Cost of takeover without activist	0.151	(0.011)
$c_m(\bar{v})$	Cost of takeover with activist	0.135	(0.047)
ξ	Activist's entry cost	0.030	(0.004)

Activists have private info on projects (acquirers) but the signal-to-noise ratio is 14.6% (3.2%)

Activists significantly reduce the turnover (takeover) cost by 22% (11%)

The expected net return to the activist is 1.1%

Roadmap

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Substitutes versus complements

<i>All values expressed as percentages</i>	No activism	Substitutes: Only effort incentives	Complements: Only takeover facilitation	Baseline
<u>Takeover</u>				
Frequency of takeover	4.19	4.16	4.22	4.19
Change from no activism		-0.79	0.64	-0.05
Frequency of takeover activist	–	11.57	14.00	12.46
Bid premium	36.92	36.98	36.69	36.70
Change from no activism		0.18	-0.63	-0.60
<u>Turnover and effort</u>				
Frequency of turnover	8.14	8.49	8.15	8.50
Change from no activism		4.35	0.10	4.50
Frequency of turnover activist	–	19.00	12.07	19.00
Productivity due to effort				
Change from no activism		1.27	0.03	1.31
<u>Activism and value</u>				
Frequency of activist entry	0.00	3.88	2.93	3.94
Target shareholder value				
Change from no activism		0.33	0.00	0.33

Activism crowds out takeovers but its effort incentive provision **benefits shareholders**

Activism increases the takeover likelihood but this provides a **little benefit to shareholders**

The threat of activism and takeovers

	(1)	(2)	(3)
<i>Percent change from baseline</i>	No activism	No takeovers	No activism & no takeovers
<u>Takeover</u>			
Frequency of takeover	0.05	-100.00	-100.00
Frequency of takeover activist	–	-100.00	–
Bid premium	0.60	–	–
<u>Turnover and effort</u>			
Frequency of turnover	-4.30	11.49	4.64
Frequency of turnover activist	–	69.23	–
Productivity due to effort	-1.29	2.89	1.12
<u>Activism and value</u>			
Frequency of activist entry	-100.00	-13.72	-100.00
Target shareholder value	-0.32	-26.64	-27.31

Activism has a positive effect on managerial effort

Both the threat of activism and the threat of takeovers add shareholder value

Activist return decomposition

	(1)	(2)
	Percent Return	Percent of Total Return
Selection effect	3.69	90.8%
Project quality (s_p)	0.59	14.4%
Synergy quality (s_w)	3.10	76.3%
Treatment effect	0.38	9.3%
Effort incentives ($c_f(v)$)	0.36	8.8%
Takeover facilitation ($c_m(v)$)	0.02	0.5%
Total activist announcement return	4.07	100.0

Most of the returns around an activist's entry announcement are due to the **selection effect**
Information about potential takeovers appears to be the **key driver** of this result

Conclusion

Both activism and takeover can serve to discipline management

Activism can also attract acquirers and facilitate takeovers

We build a model to quantify the role of activism in overcoming agency conflicts and takeovers

We find that activism significantly increases effort but slightly reduces takeover activity

The threat of activism and takeovers benefit shareholders as well